

CloserNegotiator — HAN 2026 Agent Report

Human-Agent Negotiation League (HAN) · ANAC 2026

1. Overview

CloserNegotiator is a **hybrid** human-agent negotiator. A deterministic strategy engine owns every *decision* (what to offer, what to accept); the competition LLM (qwen3:4b-instruct , used unchanged as required) owns only the *natural language* that accompanies each move. This separation is deliberate: decision quality must be robust and predictable, while the text channel is where a human is informed, anchored, and persuaded. Because the LLM never decides an offer or acceptance, latency or malformed output can never produce an irrational move — the engine always falls back to a rational action and a deterministic message.

The design targets the two additive components of the HAN score — **advantage** (utility of the agreement above the reservation value) and the **concealment / modelling** point — and was tuned against real leaderboard feedback, not just local simulation.

2. Bidding Strategy (deterministic engine)

- **Anchor.** The opening offer is our maximum-utility outcome, framed via the text channel as a *phantom anchor* — already a concession in the human's favour — which research shows is as effective as a raw anchor but far less likely to be rejected as greedy.
- **Adaptive concession.** Target utility follows $u(t) = u_{\max} - (u_{\max} - \text{floor}) \cdot t^e$, with the exponent **modulated by the opponent's observed behaviour**: against a conceding partner we hold firm and extract; against a tough one we concede sooner to secure a deal. A **stall detector** notices when the human has stopped improving their offer and accelerates to close before they walk away.
- **Floor.** $\text{floor} = \max(\text{reservation}, \text{floor_frac} \cdot u_{\max})$. Live feedback showed a high floor is too greedy for the human population (causing no-deals that tanked our mean), so the floor is set to a deal-closing level above the reservation value but below a fair split.
- **Pareto-aware offers.** Among outcomes meeting our target we propose the one most favourable to the opponent (per our opponent model) — costing us nothing yet maximising acceptance.

3. Opponent Modelling and Concealment

A human supplies no model of us, so any model on our side earns the larger share of the modelling point. We maintain a lightweight **frequency model**: per issue we count the values the opponent proposes and build a `LinearAdditiveUtilityFunction` from the normalised frequencies, exposed through `opponent_ufun`, updated every round, and fully exception-

guarded. **Concealment:** our offers are driven only by our own utility target and the accompanying text is forbidden from revealing which issues matter most to us, masking our true priority ordering behind a moving but uninformative offer stream.

4. Acceptance Strategy

1. **ACNext** — accept any offer at least as good as what we would propose next.
2. **Satisficing** — accept a genuinely good-enough offer immediately rather than holding out, because a human rewards an agent that says yes to a fair deal and may walk if pushed.
3. **Endgame / last-ditch capture** — near the deadline accept anything comfortably above (and, in the final moments, anything strictly above) the reservation value, so we never time out into a no-deal.

5. Persuasion (the LLM text channel)

The prompt makes the LLM a *persuader*, not a decision-maker: build brief rapport; present every offer as fair; justify with a concrete reason; frame concessions as goodwill; invoke reciprocity; apply gentle loss-aversion at the close; keep our priorities private; be gracious on acceptance — all in one or two short, human-sounding sentences. Tactics are **phase-aware** (opening / middle / closing) and the close uses a reciprocal commitment ask plus a gracious tone to reduce backout.

Latency safety (critical). Calling the LLM on every move once exceeded the per-run time limit. We therefore use a **smart, bounded budget**: the LLM is spent only on high-impact moments (opening anchor, the close, replies to a human objection), with a hard cap and a 12-second per-call ceiling; every other move uses an instant deterministic fair message. This keeps full persuasive power against a talking human while guaranteeing we never time out.

6. Robustness

All official scenario types (continuous, cardinal, ordinal issues) run without error. Decisions are independent of LLM availability; the opponent model is fully guarded; large outcome spaces are sampled and handled in milliseconds; the agent never accepts a below-reservation deal and never throws. It holds no state between negotiations, as required.

7. Empirical Validation

The agent was developed against (a) a diverse pool of automated negotiators, (b) a custom *behavioural-human* simulator modelling ultimatum rejection, impatience, greed-disengagement and reciprocity, and most importantly (c) the **live HAN leaderboard**. Live feedback drove the two biggest design decisions: removing per-move LLM latency (which had been silently causing time-outs) and lowering the floor to a deal-closing level (a too-greedy floor was costing us no-deals). These changes moved the agent from the bottom of

the board into the top tier; ongoing tuning of the persuasion budget and floor continues against real opponents.